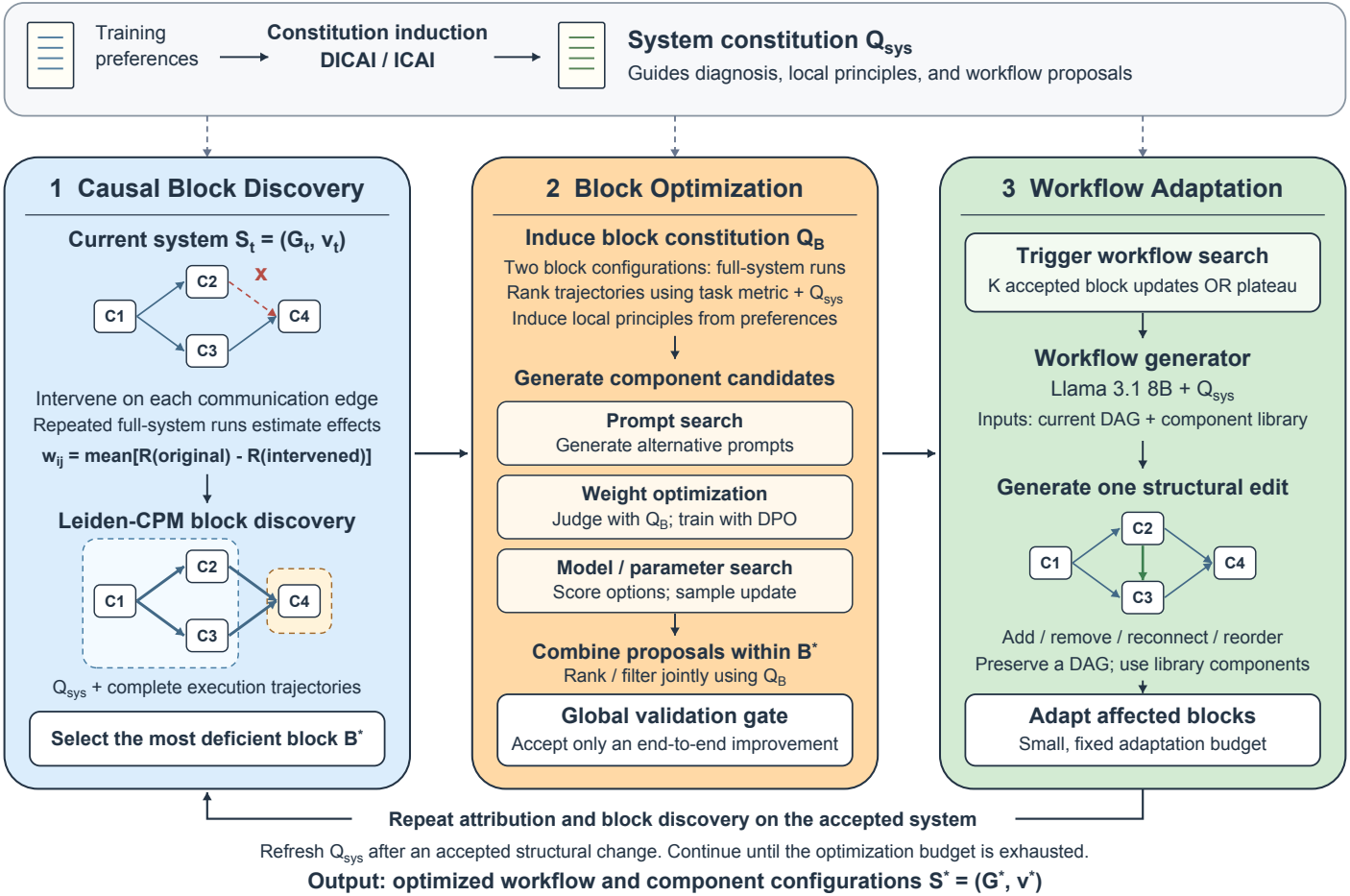

Training preferences
Constitution induction
DICAI / ICAI
System constitution Q_{sys}
Guides diagnosis, local principles, and workflow proposals
1 Causal Block Discovery
Current system $S_t = (G_t, v_t)$
C1
C2
C3
C4
X
Intervene on each communication edge
Repeated full-system runs estimate effects
w_{ij} = mean[R(original) - R(intervened)]
Leiden-CPM block discovery
Q_{sys} + complete execution trajectories
Select the most deficient block B^*
2 Block Optimization
Induce block constitution Q_B
Two block configurations: full-system runs
Rank trajectories using task metric + Q_{sys}
Induce local principles from preferences
Generate component candidates
Prompt search
Generate alternative prompts
Weight optimization
Judge with Q_B; train with DPO
Model / parameter search
Score options; sample update
Combine proposals within B^*
Rank / filter jointly using Q_B
Global validation gate
Accept only an end-to-end improvement
3 Workflow Adaptation
Trigger workflow search
K accepted block updates OR plateau
Workflow generator
Llama 3.1 8B + Q_{sys}
Inputs: current DAG + component library
Generate one structural edit
Add / remove / reconnect / reorder
Preserve a DAG; use library components
Adapt affected blocks
Small, fixed adaptation budget
Repeat attribution and block discovery on the accepted system
Refresh Q_{sys} after an accepted structural change. Continue until the optimization budget is exhausted.
Output: optimized workflow and component configurations $S^* = (G^*, v^*)$